

Kadison's similarity theorem through uniform derivation estimates

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Abstract

We prove that every bounded complex-linear unital algebra homomorphism from a unital complex C^* -algebra into the bounded operators on an arbitrary Hilbert space is similar to a $*$ -homomorphism. This resolves Kadison's similarity conjecture positively. We also obtain one universal hyperreflexivity constant for all unital von Neumann algebras on arbitrary complex Hilbert spaces.

1 Introduction

Kadison's similarity problem asks whether the adjoint operation of a C^* -algebra is respected, after a bounded change of Hilbert space coordinates, by every bounded representation of its underlying algebra. Kadison formulated the problem in 1955 in his study of the orthogonalization of operator representations [15]. Its force lies in the absence of any assumed relation between the algebraic representation and the involution: multiplicativity and boundedness alone must produce one equivalent Hilbert space inner product for which all represented adjoints are correct. This paper resolves the problem positively in its unrestricted bounded unital form.

Theorem 1.1 (Similarity). *Let A be any unital complex C^* -algebra, let H be any complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ be a bounded, complex-linear, unital algebra homomorphism. There exists an operator $S \in \mathcal{B}(H)$, invertible with bounded inverse, such that*

$$\rho(a) = S\pi(a)S^{-1} \quad (a \in A)$$

is a $$ -homomorphism. In particular, $\rho(a^*) = \rho(a)^*$ for every $a \in A$.*

The theorem imposes no separability, nuclearity, finite generation, faithfulness, or normality condition. The similarity may depend on the whole homomorphism. The result concerns the existence of this single bounded invertible operator, and does not require an a priori bound for its condition number.

1.1 Historical context

The nuclear case was established by Bunce and Christensen [2, 5]; see also the precise quantitative comparison in Pisier [22, Theorem 1]. Haagerup's theorem that every nuclear C^* -algebra is amenable [13, Theorem 3.1] connects this case with the cohomology of operator algebras. Complete boundedness provided the principal analytic framework for the general similarity problem. Haagerup solved the cyclic representation case and identified complete boundedness as the criterion for similarity to a $*$ -representation [12]. He also proved the similarity property for C^* -algebras without tracial states [12, Proposition 1.8], and Christensen established it for type II_1 factors with property Γ [6]; see also Ozawa [18, Section 1.2, Theorem 1.3]. These cases explain the role of the finite-factor argument below, where the estimate must be uniform in the factor and its representation multiplicity.

Christensen established the corresponding criterion for derivations: a derivation into the ambient bounded operators is inner precisely when it is completely bounded [4]. Paulsen's similarity theorem extends the homomorphism criterion to unital operator algebras and gives the optimal condition number [19, p. 226]. Kirchberg then proved that, for each unital C^* -algebra, the similarity property is equivalent to innerness of derivations in every faithful $*$ -representation [16]; we use the explicit per-algebra formulation in Ozawa [18, Theorem 1.1]. Ringrose's automatic continuity theorem allows this derivation property to be stated using bounded derivations [27].

A von Neumann algebra is *hyperreflexive* if the distance from any operator to the algebra is bounded by a fixed multiple of the supremum of its off-diagonal compression norms over the algebra's invariant subspaces. More recently, Eleftherakis and Paulsen [9, Theorem 2.4] characterized a positive answer to the similarity problem by preservation of hyperreflexivity when adjoining a projection: whenever $M \subset \mathcal{B}(H)$ is a hyperreflexive von Neumann algebra and $q \in \mathcal{B}(H)$ is a projection, $W^*(M, q)$ is hyperreflexive. Their result is an equivalence with this extension property and with the assertion that every von Neumann algebra is hyperreflexive. Corollary 1.3 below makes the latter consequence quantitative with one universal constant.

Pisier organized quantitative similarity estimates through the similarity degree and its interpretation as the length of matrix factorizations [20, 21]. His characterization of nuclearity by a similarity exponent below three distinguishes the strength of the nuclear case [22, Theorem 1]. These results concern quantitative bounds for homomorphisms. The proof below is organized around a uniform bound for derivations, and concludes with existence of a similarity as stated in Theorem 1.1.

The companion result on discrete groups identifies unitarizability with amenability [17, Theorem 1.1]. Unlike the homomorphisms in Theorem 1.1, a uniformly bounded group representation need not have a bounded extension to the full or reduced group C^* -algebra; the automatic extension is to $\ell^1(G)$.

1.2 The uniform estimate and proof strategy

An inner derivation is a map $a \mapsto [Y, a]$. Complete boundedness means uniform boundedness of all entrywise finite matrix amplifications. Our main estimate controls those amplifications of an inner derivation by its norm at the scalar level. For a represented unital von Neumann algebra $P \subset \mathcal{B}(K)$ and $Y \in \mathcal{B}(K)$, put

$$g_P(Y) = \sup_{a \in P, \|a\| \leq 1} \|Ya - aY\|,$$

and write $Y^{(h)}$ for the diagonal amplification of Y on K^h .

Theorem 1.2 (Uniform commutator estimate). *There is an absolute constant $C < \infty$ such that, for every complex Hilbert space K , every unital von Neumann algebra $P \subset \mathcal{B}(K)$, every $Y \in \mathcal{B}(K)$, every integer $h \geq 1$, and every $X \in M_h(P)$,*

$$\|[Y^{(h)}, X]\| \leq C g_P(Y) \|X\|. \quad (1)$$

Here $M_h(P)$ acts naturally on K^h , and $[T, U] = TU - UT$.

Corollary 1.3 (Universal hyperreflexivity). *Let H be any complex Hilbert space and let $M \subset \mathcal{B}(H)$ be a unital von Neumann algebra. For $T \in \mathcal{B}(H)$, set*

$$\alpha_M(T) = \sup\{\|(1 - e)Te\| : e \in M', e = e^* = e^2\}.$$

Then, with the absolute constant C from Theorem 1.2,

$$\text{dist}(T, M) \leq 2C \alpha_M(T) \quad (T \in \mathcal{B}(H)). \quad (2)$$

Thus every such M is hyperreflexive with constant at most $2C$, independently of the algebra, its representation, and the Hilbert space. No optimality of this constant is asserted.

Applying the commutator estimate to M' gives this distance bound through Arveson's commutant-distance formula. The proof is given in Section 7.5.

The difficult part is obtaining a constant independent of matrix size for finite factors. Section 2 first combines Dickson's row estimate [8, Lemma 2.9] with its column counterpart to implement rectangular derivations with cyclic domain. Here rectangular derivations take values in operators between two representation spaces. The implementation norm is controlled by the derivation norm with an absolute constant.

Sections 3–5 use this cyclic-domain estimate in a tracial free product $\mathcal{L} = D * A_0$, where D is a II_1 factor generated by increasing full matrix algebras and A_0 is a finite von Neumann algebra with a faithful normal tracial state. The input is an operator G on $L^2(\mathcal{L})$ commuting with left multiplication by D , whose commutators with elements of a corner $e\mathcal{L}e$ have norm at most ϵ times the element norm. Here $e \in D$ is a projection of trace $1/N$, for an integer $N \geq 2$. The output bounds the compression to $L^2(A_0)$ of its commutator with any trace-zero self-adjoint unitary in A_0 , by an absolute multiple of ϵ independent of the corner trace. This is Theorem 3.1.

The reduced-word model belongs to the free-product framework developed by Ching and Voiculescu [3, 29]. Its role here is to turn the corner hypothesis into a statement about operators on Hilbert tensor products.

In Section 3, averaging over the finite matrix unitary groups reveals the structure of the operator on reduced words. Finite-dimensional Schur–Weyl duality supplies invariant cycle-trace coefficients; the endpoint L^1 bound and boundedness on Hilbert tensor products eliminate all contractions except permutations between equally many source and test letters. At each word level, repeated-letter tensors consequently carry an action on their remaining factors that is independent of the repeated D -letter. Section 4 approximates G , on each sufficiently small right support, by a bounded right multiplier. This propagates the repeated-letter action through arbitrarily long words with a uniform error. Section 5 compresses a multiplication operator built from two free trace-zero self-adjoint unitaries. The resulting Hankel matrix has norm bounded below as the corner trace tends to zero. The trace factors in the upper and lower bounds cancel, completing the free-product corner estimate.

Section 6 uses Popa's relative independence theorem [25, Theorem 0.1(a)] to place this free product inside the tracial ultrapower of an arbitrary separable-predual II_1 factor after matrix amplification. The irreducible-inclusion case already appears in Popa's free-independent sequence construction [24, Theorem 2.1 and proof of Corollary 2.2]. The operator under study is passed through a Hilbert space ultrapower and compressed to the tracial L^2 -space, preserving the necessary commutator coefficients. Section 7 extends the estimate to arbitrary normal factor representations, then uses central decomposition and separable reducing subspaces to obtain Theorem 1.2. The resulting constant is numerical and independent of matrix size, representation, and Hilbert space. Finally, using the cyclic-domain estimate again, finite collections of cyclic columns of a $*$ -representation implement a given derivation locally. Their scalar commutator norms are uniformly bounded, so (1) gives complete boundedness of the full derivation. Paulsen's theorem implements it, and Kirchberg's equivalence proves Theorem 1.1.

2 Rows, cyclic modules, and derivations

This section supplies two inputs to the proof: a uniform implementer for a rectangular derivation with cyclic domain, and the per-algebra derivation criterion that will conclude similarity. The cyclic estimate will first be used to extend control from a small free-product corner to an entire right-supported subspace.

All Hilbert spaces and algebras are complex, and inner products are linear in the first variable. All algebra homomorphisms are complex-linear, and all $*$ -representations are unital unless stated otherwise. No separability or faithfulness is implicit. We write E_1 for the closed unit ball of a normed space E . For a unital self-adjoint operator algebra $P \subset \mathcal{B}(K)$ and $Y \in \mathcal{B}(K)$, set

$$g_P(Y) = \sup_{a \in P_1} \|Ya - aY\|. \quad (3)$$

The same notation will be used when P is a specified represented copy of an abstract algebra.

For a linear map $u : A \rightarrow \mathcal{B}(K, F)$, its amplification $u_{p,q}$ acts entrywise on $M_{p,q}(A)$, with values in $\mathcal{B}(K^q, F^p)$. Rectangular matrix norms on A are the usual C^* -matrix norms. We abbreviate $u_{h,h}$ to u_h , and set

$$\|u\|_{\text{cb}} = \sup_{h \geq 1} \|u_h\|, \quad \|u\|_{\text{row}} = \sup_{h \geq 1} \|u_{1,h}\|, \quad \|u\|_{\text{col}} = \sup_{h \geq 1} \|u_{h,1}\|.$$

A map is completely bounded when its cb norm is finite. For an operator Y , the notation $Y^{(h)}$ means the diagonal operator on h copies of its Hilbert space.

If (P, τ) is a finite von Neumann algebra with a faithful normal tracial state, $L^2(P) = L^2(P, \tau)$ is the completion of P for $\langle x, y \rangle = \tau(y^*x)$. We write $\|\cdot\|_2$ for this norm, and $\|x\|_1 = \tau(|x|)$ for $x \in P$. Left and right multiplication are denoted by L_a and R_a , with $R_a \xi = \xi a$; thus $R_a R_b = R_{ba}$. The tracial commutation theorem identifies $L(P)'$ with $\{R_a : a \in P\}$. The support notation for an L^2 -vector is understood through its affiliated operator: if $\xi = v|\xi|$ is its polar decomposition, its left and right support projections are vv^* and v^*v , respectively. They are equivalent in P , and in particular have the same trace. For a projection $r \in P$, the condition that R_b vanish on $L^2(P)(1 - r)$ is exactly $b = rb$. We use these standard facts, and standard-form theory for von Neumann algebras, without separability restrictions [11, 28].

2.1 Uniform row and column estimates

Dickson's argument derives the row estimate from Haagerup's little Grothendieck inequality and credits Christensen with the estimate in this form [8, Lemmas 2.8–2.9]; this is part of the factorization approach to similarity developed by Pisier [20, 21]. The estimate states that every bounded unital homomorphism $\varphi : A \rightarrow \mathcal{B}(K)$, for any unital C^* -algebra A , satisfies

$$\|\varphi\|_{\text{row}} \leq \sqrt{2} \|\varphi\|^2 \quad [8, \text{Lemma 2.9, Equation (19)}]. \quad (4)$$

The map $\varphi^\sharp(a) = \varphi(a^*)^*$ is complex-linear and has the same norm as φ . It is again a homomorphism, since

$$\varphi^\sharp(ab) = (\varphi(b^*)\varphi(a^*))^* = \varphi^\sharp(a)\varphi^\sharp(b).$$

Apply (4) to $\varphi^\sharp$ on the row $[a_1^*, \dots, a_h^*]$, and take the rectangular adjoint. This gives the same bound for $\|\varphi\|_{\text{col}}$. Consequently we fix the absolute constant

$$c_{\text{row}} = 4\sqrt{2}, \quad (5)$$

which bounds both norms whenever $\|\varphi\| \leq 2$.

Let A be a unital C^* -algebra. A *rectangular derivation* for $*$ -representations $\sigma : A \rightarrow \mathcal{B}(K)$ and $\lambda : A \rightarrow \mathcal{B}(F)$ is a linear map $\Delta : A \rightarrow \mathcal{B}(K, F)$ satisfying

$$\Delta(ab) = \Delta(a)\sigma(b) + \lambda(a)\Delta(b) \quad (a, b \in A). \quad (6)$$

The derivation identity and unitality imply $\Delta(1) = 0$.

Lemma 2.1 (Row and column bounds). *Every bounded rectangular derivation satisfies*

$$\max\{\|\Delta\|_{\text{row}}, \|\Delta\|_{\text{col}}\} \leq c_{\text{row}}\|\Delta\|. \quad (7)$$

Proof. The assertion is immediate when $\Delta = 0$. Otherwise put $d = \|\Delta\| > 0$, and consider the upper triangular map

$$\Phi(a) = \begin{pmatrix} \lambda(a) & d^{-1}\Delta(a) \\ 0 & \sigma(a) \end{pmatrix} \in \mathcal{B}(F \oplus K). \quad (8)$$

It is unital and multiplicative by (6). Its diagonal map is contractive, and its off-diagonal map has norm one; hence $\|\Phi\| \leq 2$. The off-diagonal rectangular compression of $\Phi_{p,q}$, from K^q to F^p , is $d^{-1}\Delta_{p,q}$. The row and column estimates for Φ give (7). $\square$

2.2 Completely bounded implementation

We use the quantitative completely bounded similarity theorem of Paulsen: if a unital homomorphism $\varphi : A \rightarrow \mathcal{B}(H)$ is completely bounded, there is an invertible $S \in \mathcal{B}(H)$ such that $S\varphi(\cdot)S^{-1}$ is completely contractive and $\|S\|\|S^{-1}\| \leq \|\varphi\|_{\text{cb}}$ [19, p. 226]. For a unital C^* -algebra the resulting homomorphism is a $*$ -homomorphism: a unitary and its inverse are sent to contractions, so the image is a unitary; unitaries linearly span the algebra. For the later cyclic-domain estimate, set

$$c_0 = (1 + 4c_{\text{row}})^2. \quad (9)$$

Lemma 2.2 (Completely bounded rectangular derivations). *Let σ, λ, Δ satisfy (6). If Δ is completely bounded, then some $V \in \mathcal{B}(K, F)$ satisfies*

$$\Delta(a) = V\sigma(a) - \lambda(a)V \quad (a \in A). \quad (10)$$

If $d = \|\Delta\| > 0$, one may choose

$$\|V\| \leq d(1 + \|\Delta\|_{\text{cb}}/d)^2.$$

Proof. We use the triangular-metric argument recalled in Ozawa [18, Lemma 1.2], allowing different domain and codomain representations and keeping track of the implementer norm. For $\Delta = 0$, take $V = 0$. Otherwise the map Φ in (8) is completely bounded, with

$$\|\Phi\|_{\text{cb}} \leq 1 + \|\Delta\|_{\text{cb}}/d =: G.$$

Choose the similarity S supplied by Paulsen's theorem, and put $D = S^*S$. The fact that $S\Phi(\cdot)S^{-1}$ preserves adjoints is equivalent to

$$D\Phi(a) = \Phi(a^*)^*D \quad (a \in A). \quad (11)$$

Write $D = (D_{ij})_{i,j=1}^2$ relative to $F \oplus K$. Its first row in (11) gives

$$\begin{aligned} D_{11}\lambda(a) &= \lambda(a)D_{11}, \\ d^{-1}D_{11}\Delta(a) + D_{12}\sigma(a) &= \lambda(a)D_{12}. \end{aligned} \quad (12)$$

Since $D \geq \|D^{-1}\|^{-1}I$, its compression D_{11} is positive invertible on F , with $\|D_{11}^{-1}\| \leq \|D^{-1}\|$. Its inverse also commutes with $\lambda(A)$. Hence

$$V = -dD_{11}^{-1}D_{12}$$

satisfies (10), by (12), and

$$\|V\| \leq d \|D^{-1}\| \|D\| = d (\|S^{-1}\| \|S\|)^2 \leq dG^2.$$

If either Hilbert space is zero, $\Delta = 0$, so it has already been covered. $\square$

2.3 Cyclic domains

The cyclic hypothesis needed below concerns only the domain $*$ -representation. The output representation may have arbitrary multiplicity.

Lemma 2.3 (Column factorization in a cyclic representation). *Let $P \subset \mathcal{B}(K)$ be a unital von Neumann algebra with a cyclic vector. For every integer $h \geq 1$ and every unit vector $\xi = (\xi_1, \dots, \xi_h) \in K^h$, there are a unit vector $\zeta \in K$ and a contraction $v = (v_i) \in M_{h,1}(P)$ such that $\xi_i = v_i \zeta$ for every i .*

Proof. Put $N = P'$, and choose a cyclic vector Ω for P . It separates N : if $x \in N$ and $x\Omega = 0$, then $xa\Omega = ax\Omega = 0$ for every $a \in P$, whence $x = 0$. Let $L = \overline{N\Omega} \subset K$. This subspace reduces N , and restriction $r : N \rightarrow \mathcal{B}(L)$ is faithful and normal. Indeed, an element in its kernel annihilates Ω ; normality follows from compression to a reducing subspace. Thus it is a normal $*$ -isomorphism onto the von Neumann algebra $r(N)$, and its inverse is normal. The vector Ω is cyclic and separating for $r(N)$.

The standard-form theorem for a representation with a cyclic separating vector realizes every normal positive functional on $r(N)$ as a vector functional, with its vector in the natural cone [11, Lemma 2.10(1), pp. 278–279]. Apply it, through the normal isomorphism r , to

$$\omega(x) = \sum_{i=1}^h \langle x\xi_i, \xi_i \rangle, \quad x \in N.$$

We obtain $\zeta \in L \subset K$ such that $\langle x\zeta, \zeta \rangle = \omega(x)$ for every $x \in N$. In particular $\|\zeta\|^2 = \omega(1) = 1$. The assignment

$$x\zeta \mapsto (x\xi_1, \dots, x\xi_h), \quad x \in N,$$

is isometric, since both squared norms equal $\omega(x^*x)$. It extends to an isometry from $\overline{N\zeta}$ into K^h . Extend it by zero on $\overline{N\zeta}^\perp$, obtaining a contraction $v : K \rightarrow K^h$. The subspace $\overline{N\zeta}$ reduces N , and the isometry intertwines the N -actions. Therefore $vx = x^{(h)}v$ for every $x \in N$. Each coordinate v_i lies in $N' = P$, and $v\zeta = \xi$, as required. All these constructions, including the standard-form assertion, are valid on nonseparable Hilbert spaces. $\square$

Lemma 2.4 (Implementation on a cyclic domain). *Let A be a unital C^* -algebra, let $\sigma : A \rightarrow \mathcal{B}(K)$ and $\lambda : A \rightarrow \mathcal{B}(F)$ be $*$ -representations, and suppose that σ has a cyclic vector. Every bounded rectangular derivation $\Delta : A \rightarrow \mathcal{B}(K, F)$ is completely bounded, with*

$$\|\Delta\|_{\text{cb}} \leq 4c_{\text{row}}\|\Delta\|. \quad (13)$$

Moreover there is $V \in \mathcal{B}(K, F)$ such that

$$\Delta(a) = V\sigma(a) - \lambda(a)V \quad (a \in A), \quad \|V\| \leq c_0\|\Delta\|. \quad (14)$$

Proof. Zero spaces and the zero derivation are immediate, so assume they do not occur. Put $P = \sigma(A)''$. Fix $h \geq 1$, $X \in M_h(A)_1$, and a unit vector $\xi \in K^h$. Lemma 2.3 gives a unit vector $\zeta \in K$ and a contraction $v \in M_{h,1}(P)$ with $v\zeta = \xi$.

Matrix Kaplansky density gives a net of column contractions $z_\alpha \in M_{h,1}(\sigma(A))$ converging strongly to v . To see the rectangular assertion directly, regard v as the first column of a matrix in $M_h(P)$, with all other columns zero. Approximate that matrix strongly by contractions in $M_h(\sigma(A))$, and compress to its first input coordinate. The $*$ -image $\sigma(A)$ is the C^* -quotient $A/\ker \sigma$, and the same quotient identification holds at each square matrix level. Consequently each z_α lifts to $w_\alpha \in M_{h,1}(A)$ with

$$\|w_\alpha\| \leq 2, \quad \sigma_{h,1}(w_\alpha) = z_\alpha, \quad \sigma_{h,1}(w_\alpha)\zeta \longrightarrow \xi. \quad (15)$$

For example, first lift the square matrix whose first column is z_α and whose other columns vanish to a matrix of norm less than 2, then compress its first column. Only fixed slack in the quotient norm is being used.

Entrywise application of (6) gives

$$\Delta_h(X)\sigma_{h,1}(w_\alpha) = \Delta_{h,1}(Xw_\alpha) - \lambda_h(X)\Delta_{h,1}(w_\alpha).$$

Since λ is completely contractive, Lemma 2.1 and (15) imply

$$\|\Delta_h(X)\sigma_{h,1}(w_\alpha)\zeta\| \leq c_{\text{row}}\|\Delta\|(\|Xw_\alpha\| + \|\lambda_h(X)\|\|w_\alpha\|) \leq 4c_{\text{row}}\|\Delta\|.$$

At the fixed finite level h , $\Delta_h(X)$ is already a bounded operator. We may therefore take the limit in (15), obtaining the same bound on $\|\Delta_h(X)\xi\|$. Taking the supremum over ξ, X, h proves (13). The quotient argument did not require Δ to vanish on $\ker \sigma$; its identity was applied to the chosen lifts in A . Lemma 2.2 and (9) now give (14). $\square$

2.4 The derivation criterion for similarity

For the last step of the proof we use Kirchberg's equivalence in its per-algebra form. We state the quantifiers because the final $*$ -representation used to test derivations is independent of the bounded homomorphism in the similarity problem.

Theorem 2.5 (Kirchberg's similarity–derivation equivalence). *For a unital C^* -algebra A , the following properties are equivalent.*

1. *Every bounded unital homomorphism $\pi : A \rightarrow \mathcal{B}(H)$, on any Hilbert space H , is similar by a bounded invertible operator to a $*$ -homomorphism.*
2. *For every faithful unital $*$ -representation $\sigma : A \rightarrow \mathcal{B}(K)$, on any Hilbert space K , every bounded linear map $\Delta : A \rightarrow \mathcal{B}(K)$ satisfying*

$$\Delta(ab) = \Delta(a)\sigma(b) + \sigma(a)\Delta(b)$$

has the form $\Delta(a) = V\sigma(a) - \sigma(a)V$ for some $V \in \mathcal{B}(K)$.

This is the theorem of Kirchberg [16]; the exact formulation with every faithful representation is stated in Ozawa [18, Theorem 1.1]. The usual derivation property is stated without a continuity hypothesis. Ringrose's automatic continuity theorem for derivations from C^* -algebras into Banach bimodules [27, Theorem 2, p. 434] makes it equivalent to the bounded version above. The triangular metric computation in Lemma 2.2 uses only Paulsen's established completely bounded theorem; the unrestricted similarity conclusion will follow from Theorem 2.5 after all bounded derivations have been treated.

3 Free-product geometry and invariant tensors

Let D be a II_1 factor generated as a von Neumann algebra by an increasing sequence (D_j) of unital full matrix subalgebras whose sizes tend to infinity. Let A_0 be a finite von Neumann algebra with a faithful normal tracial state, and form the tracial von Neumann free product

$$\mathcal{L} = D * A_0.$$

We use τ for the free-product trace and its restrictions. Freeness here is scalar tracial freeness: the trace of an alternating product of centered letters from the two algebras is zero. Its Hilbert-space consequence is the orthogonal decomposition into reduced words, with each reduced-word space isometric to the corresponding Hilbert tensor product. This model comes from the free-product framework of Ching [3] and Voiculescu [29]. The corner estimate below is proved here from that word decomposition.

This section and the next two prove the following estimate. Its constant does not depend on the trace of the corner.

Theorem 3.1 (Free-product corner estimate). *Let $\mathcal{L} = D * A_0$ be as above, let $t = 1/N$ for an integer $N \geq 2$, and suppose that $G \in \mathcal{B}(L^2(\mathcal{L}))$ commutes with L_D . Suppose that $\epsilon \geq 0$ and*

$$\|[G, L_x]\| \leq \epsilon \|x\| \quad (x \in e\mathcal{L}e, e \in D \text{ a projection with } \tau(e) = t). \quad (16)$$

If $s = s^* = s^{-1} \in A_0$ and $\tau(s) = 0$, then

$$\left\| P_{L^2(A_0)}[G, L_s] \right\|_{L^2(A_0)} \leq c_1 \epsilon, \quad c_1 = 6\pi^2(1 + 4c_0). \quad (17)$$

Since equal-trace projections in D are unitarily conjugate, commutation with L_D makes it equivalent to assume (16) for one such projection. Moreover, for $x \in e\mathcal{L}e$ the commutator in that equation is supported on $eL^2(\mathcal{L})$. The proof of the theorem is completed after the Fourier compression calculation in Section 5.

3.1 Factoriality and symmetrization

Lemma 3.2. *One has $D' \cap \mathcal{L} = \mathbb{C}1$. In particular, $\mathcal{L}$ is a II_1 factor.*

Proof. Put $E = L^2(D)$, $E^\circ = E \ominus \mathbb{C}1$, and $B^\circ = L^2(A_0) \ominus \mathbb{C}1$. Regrouping reduced words by their first and last A_0 -letters gives the D - D bimodule decomposition

$$L^2(\mathcal{L}) \ominus L^2(D) \cong E \otimes \mathcal{V} \otimes E, \quad \mathcal{V} = \bigoplus_{n \geq 1} B^\circ \otimes (E^\circ \otimes B^\circ)^{\otimes(n-1)}.$$

The endpoints in E are unrestricted; expanding each as $\mathbb{C}1 \oplus E^\circ$ recovers every reduced word containing an A_0 -letter exactly once. Left and right D act on the first and last factors, respectively.

Choose a Haar unitary v in a diffuse abelian subalgebra of D and put $u_n = v^n$. These unitaries converge ultraweakly to zero in D . Indeed, the restriction of any normal functional to $W^*(v) \cong L^\infty(\mathbb{T})$ has an $L^1(\mathbb{T})$ density, whose Fourier coefficients tend to zero. Consequently L_{u_n} and $R_{u_n^*}$ converge weakly to zero on E : first test against bounded vectors in D , and then use their L^2 density. On elementary tensors the matrix coefficients of

$$L_{u_n} R_{u_n^*} = L_{u_n} \otimes I_{\mathcal{V}} \otimes R_{u_n^*}$$

therefore tend to zero. The same holds on the whole complement by density and uniform boundedness. A D -central vector in this complement is fixed by all these operators, and hence is zero.

If $z \in D' \cap \mathcal{L}$, its component perpendicular to $L^2(D)$ is D -central, because the displayed decomposition is a bimodule decomposition. Thus $z = E_D(z) \in D$, where E_D is the trace-preserving conditional expectation. It follows that $z \in Z(D) = \mathbb{C}1$. The center of $\mathcal{L}$ is contained in this relative commutant. Its faithful finite trace and its diffuse subfactor D then make $\mathcal{L}$ a II_1 factor. $\square$

For $u \in \mathcal{U}(D)$, let θ_u be the trace-preserving free-product automorphism which equals $\text{Ad } u$ on D and is the identity on A_0 . Its Hilbert-space implementation W_u acts by θ_u on bounded words. In particular,

$$W_u L_x W_u^* = L_{\theta_u(x)}, \quad W_u \xi = \xi \quad (\xi \in L^2(A_0)).$$

Lemma 3.3 (Symmetrization). *In proving Theorem 3.1, one may suppose in addition that G commutes with W_u for every $u \in \bigcup_j \mathcal{U}(D_j)$. This replacement preserves the compressed commutator in (17) exactly, and preserves (16) for both G and G^* .*

Proof. Average the original operator over each compact matrix unitary group:

$$G_j = \int_{\mathcal{U}(D_j)} W_u G W_u^* d\mu_j(u).$$

This weak operator integral is well defined. The unitary representation $u \mapsto W_u$ is strongly continuous on each group, as is seen first on bounded reduced words and then by density. Every conjugate in the integral commutes with L_D . It also obeys (16), since θ_u permutes the D -corners of trace t . Convexity and weak operator closedness of norm balls preserve these properties under integration and under a weak operator cluster limit $\tilde{G}$ as $j \rightarrow \infty$. Such a limit exists because $\|G_j\| \leq \|G\|$. For fixed $u \in \mathcal{U}(D_i)$, all G_j with $j \geq i$ commute with W_u , by Haar invariance. Thus $\tilde{G}$ has every stated symmetry.

The operators W_u fix $L^2(A_0)$ pointwise and commute with L_s . For each conjugate, and therefore for the averages and their limit,

$$P_{L^2(A_0)}[W_u G W_u^*, L_s] \big|_{L^2(A_0)} = P_{L^2(A_0)}[G, L_s] \big|_{L^2(A_0)}.$$

Finally, $[G^*, L_x] = -[G, L_x^*]^*$, so the hypothesis for a self-adjoint corner algebra also gives its adjoint version. The same applies to $\tilde{G}$. $\square$

For the remainder of the proof of Theorem 3.1, G denotes the symmetrized operator. At the end we recover the original compressed commutator using Lemma 3.3.

3.2 The word decomposition

Set

$$E = L^2(D), \quad E^\circ = E \ominus \mathbb{C}1, \quad B = L^2(A_0), \quad B^\circ = B \ominus \mathbb{C}1, \quad J_k = (B^\circ)^{\otimes k} \otimes B.$$

We use the following ordering of the reduced-word factors:

$$L^2(\mathcal{L}) = \bigoplus_{k \geq 0} \mathcal{H}_k, \quad \mathcal{H}_k = E \otimes (E^\circ)^{\otimes k} \otimes J_k. \quad (18)$$

The tensor

$$d_0 \otimes d_1 \otimes \cdots \otimes d_k \otimes (a_1 \otimes \cdots \otimes a_k \otimes z)$$

represents the word $d_0 a_1 d_1 \cdots a_k d_k z$. Initially the letters may be taken bounded, with $d_i \in D \oplus \mathbb{C}1$ and $a_i \in A_0 \oplus \mathbb{C}1$ for $i \geq 1$, and unrestricted $d_0 \in D, z \in A_0$. Expanding the endpoints into scalar and centered parts produces all the usual reduced-word spaces exactly once. This proves both orthogonality and completeness in (18); the tensor notation then extends to arbitrary L^2 -letters by the corresponding isometries. On each level, L_D acts on the first factor, while W_u conjugates all $k+1$ factors coming from D and fixes J_k .

We will show that G preserves each word level $\mathcal{H}_k$ and, on tensors with a repeated centered D letter, changes only the J_k factor. Scalar slices against fixed vectors in the J_k and J_l factors reduce these assertions to the following intertwiner calculation. Write $C_u \xi = u \xi u^*$ on E and on its invariant subspace E° .

Lemma 3.4 (Invariant intertwiners). *Let $k, l \geq 0$, and let*

$$Y : E \otimes (E^\circ)^{\otimes k} \longrightarrow E \otimes (E^\circ)^{\otimes l}$$

be bounded. Suppose that Y intertwines the left D actions and that

$$Y C_u^{\otimes(k+1)} = C_u^{\otimes(l+1)} Y \quad (u \in \bigcup_j \mathcal{U}(D_j)).$$

Then $Y = 0$ if $k \neq l$. If $k = l$, there are scalars $(a_\sigma)_{\sigma \in S_k}$ such that

$$Y = I_E \otimes \sum_{\sigma \in S_k} a_\sigma U_\sigma, \quad U_\sigma(x_1 \otimes \cdots \otimes x_k) = x_{\sigma^{-1}(1)} \otimes \cdots \otimes x_{\sigma^{-1}(k)}.$$

For $k = 0$ the sum consists of the scalar multiple of the identity corresponding to the empty permutation.

Proof. For a vector or algebra element $x \in E$, put $x^\circ = x - \tau(x)1$. On unrestricted variables $a, x_1, \dots, x_k, z_1, \dots, z_l \in D$, define the complex multilinear form

$$F(a, x, z) = \langle Y(1 \otimes x_1^\circ \otimes \cdots \otimes x_k^\circ), a^* \otimes (z_1^*)^\circ \otimes \cdots \otimes (z_l^*)^\circ \rangle.$$

It is multilinear with our convention that inner products are linear in the first argument. Fixing the tail vectors and taking their Hilbert-space slice of Y gives an operator $T : E \rightarrow E$ commuting with L_D . The tracial commutation theorem gives $T = R_b$ for a bounded $b \in D$, with

$$\|b\| = \|T\| \leq \|Y\| \prod_i \|x_i^\circ\|_2 \prod_j \|(z_j^*)^\circ\|_2.$$

Since $F(a, x, z) = \tau(ab)$, this yields the endpoint estimate

$$|F(a, x, z)| \leq \|Y\| \|a\|_1 \prod_{i=1}^k \|x_i\|_2 \prod_{j=1}^l \|z_j\|_2. \quad (19)$$

The same slice also identifies coefficients with arbitrary bounded first endpoints: replacing $1, a^*$ by d, c replaces a by $c^* d$, because $\langle db, c \rangle = \tau(c^* db)$.

Finite matrix invariants and stable coefficients. Put $p = 1 + k + l$ and restrict all variables to a stage $D_j \cong M_m(\mathbb{C})$. The intertwining assumption says that F is invariant under simultaneous unitary conjugation. To see precisely the finite-dimensional consequence, label its variables $v_0, \dots, v_{p-1}$. Every multilinear form on M_m^p has a unique expression

$$\mathrm{Tr}(A(v_0 \otimes \cdots \otimes v_{p-1})) \quad (A \in \mathrm{End}((\mathbb{C}^m)^{\otimes p})),$$

by the nondegenerate trace pairing. Conjugation invariance is equivalent to A commuting with every $u^{\otimes p}$. Finite-dimensional Schur–Weyl duality identifies this commutant with the span of the tensor permutation operators; they are linearly independent when $m \geq p$ [10, Theorem 3.1]. Taking their trace pairings gives the cycle-trace forms

$$\Phi_\pi(v_0, \dots, v_{p-1}) = \prod_{(i_1 \dots i_r) \text{ a cycle of } \pi} \tau(v_{i_1} \dots v_{i_r}), \quad \pi \in S_p.$$

Here cycles follow the successor convention of the displayed product; reversing the convention only reindexes the permutations. The normalization from Tr to τ multiplies each form by a nonzero scalar, so these forms are also independent for $m \geq p$. Thus

$$F|_{D_j^p} = \sum_{\pi \in S_p} \lambda_{\pi,j} \Phi_\pi.$$

This identity is on *unrestricted* matrix variables: the centering occurs inside the definition of F . Under a unital inclusion of full matrix stages, normalized traces agree, and hence every Φ_π restricts to the identically defined form on the smaller stage. Uniqueness on any smaller stage of size at least p proves that $\lambda_{\pi,j}$ is independent of j from that point onward. Denote it by λ_π .

Isolating coefficients before restricting to centered tails. The invariant-theory step has expressed every coefficient through finitely many cycle-trace forms with coefficients fixed across matrix stages. We now use the endpoint bound to remove long cycles, then Hilbert-space boundedness to remove pairs within one side. Fix $\pi \in S_p$. Choose distinct matrix indices for the slots and put $v_i = e_{i,\pi(i)}$. In a nonzero product of these matrix units, the variable in slot i must be followed by the variable whose initial index is $\pi(i)$. Closing each product under its trace therefore forces exactly the cycles of π . Consequently

$$\Phi_\rho(v_0, \dots, v_{p-1}) = \begin{cases} m^{-c(\pi)}, & \rho = \pi, \\ 0, & \rho \neq \pi, \end{cases}$$

where $c(\pi)$ is the number of cycles. Every chosen matrix unit has L^1 norm m^{-1} and L^2 norm $m^{-1/2}$, including diagonal units. Applying (19) gives

$$|\lambda_\pi| m^{-c(\pi)} \leq \|Y\| m^{-1-(k+l)/2}.$$

The coefficient is fixed while the stage size tends to infinity. It follows that

$$\lambda_\pi = 0 \quad \text{if} \quad c(\pi) < 1 + \frac{k+l}{2}.$$

Now restrict the $k+l$ tail variables to centered matrices. Any form with a singleton tail cycle vanishes. For a permutation with no such singleton, let r be the length of the cycle containing the distinguished variable $a = v_0$. Its number of cycles is at most

$$1 + \frac{k+l-(r-1)}{2}.$$

If $r > 1$, this is below the required threshold. If $r = 1$, the threshold is attained only when every tail cycle has length two. Thus the only remaining terms have a as a singleton and pair all tail slots. On centered tails,

$$F(a, x, z) = \tau(a) \sum_{\mathcal{P}} \lambda_{\mathcal{P}} \prod_{\{i,j\} \in \mathcal{P}} \tau(v_i v_j),$$

where $\mathcal{P}$ ranges over pairings of the tail slots. There are no terms when the number of tail slots is odd.

Using $a = c^*d$ gives the corresponding simple-tensor coefficients with first endpoints d, c in the union of the matrix stages. These formulas extend to arbitrary endpoints in E and arbitrary centered L^2 tail letters. Indeed, that union is L^2 dense, the left side is continuous in its individual L^2 letters, and each factor on the right is continuous by $|\tau(uv)| \leq \|u\|_2\|v\|_2$. At this point the assertion is continuity of coefficients on simple tensors, not boundedness of each pairing on the completed tensor product.

Excluding pairings within one side. Choose an infinite orthonormal family (h_n) of self-adjoint centered vectors in E . For example, normalized real trigonometric functions of a Haar unitary give such a family. Then $\tau(h_n h_m) = \delta_{nm}$. Fix a pairing $\mathcal{P}$, with r_s pairs entirely in the source slots and r_t entirely in the test slots. Assign a disjoint infinite pool of the h_n to each pair. For each source–test pair, use one fixed member of its pool in both slots. For each source–source pair, sum the source tensor over M members of its pool, placing the same member in its two slots, independently for the different pairs. Form the test tensor analogously for the test–test pairs. Take both first endpoints to be 1.

In any term of the expanded coefficient every label occurs exactly twice, in the positions assigned by $\mathcal{P}$, and different pairs have different labels. Hence all other pairings contribute zero. The resulting coefficient is exactly $\lambda_{\mathcal{P}} M^{r_s+r_t}$. The source summands are orthonormal, as are the test summands, so the product of the two tensor norms is $M^{(r_s+r_t)/2}$. Boundedness of Y gives

$$|\lambda_{\mathcal{P}}| M^{r_s+r_t} \leq \|Y\| M^{(r_s+r_t)/2}.$$

Letting $M \rightarrow \infty$ proves $\lambda_{\mathcal{P}} = 0$ whenever $r_s + r_t > 0$. The argument also applies when one side has no tail slots.

All surviving pairs therefore join a source slot to a test slot, which requires $k = l$. If these numbers differ, all simple-tensor coefficients vanish and $Y = 0$. Otherwise these pairings are indexed by bijections $\sigma \in S_k$, and the coefficients are

$$\tau(c^*d) \sum_{\sigma \in S_k} a_{\sigma} \prod_{i=1}^k \tau(x_i z_{\sigma(i)}).$$

Since $\tau(x_i z_j) = \langle x_i, z_j^* \rangle$, these are precisely the coefficients of $I_E \otimes \sum_{\sigma} a_{\sigma} U_{\sigma}$. Equality against all elementary test tensors, and their density, proves the asserted operator identity. $\square$

Lemma 3.5 (Word structure). *The symmetrized operator G preserves every level $\mathcal{H}_k$ in (18). There are operators $B_k \in \mathcal{B}(J_k)$, with $\|B_k\| \leq \|G\|$, such that, for all $d_0 \in E$, $x \in E^{\circ}$, and $\alpha \in J_k$,*

$$\begin{aligned} G(d_0 \otimes x^{\otimes k} \otimes \alpha) &= d_0 \otimes x^{\otimes k} \otimes B_k \alpha, \\ G^*(d_0 \otimes x^{\otimes k} \otimes \alpha) &= d_0 \otimes x^{\otimes k} \otimes B_k^* \alpha. \end{aligned} \tag{20}$$

These are identities in the full space $L^2(\mathcal{L})$. The operators B_k are independent of the repeated letter x ; in particular, x may be any complex centered L^2 vector.

Proof. Let P_k be the level projection. It commutes with L_D and all W_u . For $\alpha \in J_k$ and $\beta \in J_l$, take the scalar slice in these factors of $P_l G P_k$ to obtain an operator

$$Y_{\alpha, \beta} : E \otimes (E^{\circ})^{\otimes k} \longrightarrow E \otimes (E^{\circ})^{\otimes l}, \quad \|Y_{\alpha, \beta}\| \leq \|G\| \|\alpha\| \|\beta\|.$$

It satisfies the hypotheses of Lemma 3.4. For $k \neq l$ every such slice vanishes, so $P_l G P_k = 0$. Thus the levels reduce G .

Fix a unit vector $h \in E^\circ$ and let $V_k : J_k \rightarrow \mathcal{H}_k$ be the isometry $V_k \alpha = 1 \otimes h^{\otimes k} \otimes \alpha$. Define $B_k = V_k^* G V_k$. On an equal-level scalar slice, the lemma gives

$$Y_{\alpha, \beta} = I_E \otimes \sum_{\sigma \in S_k} a_\sigma(\alpha, \beta) U_\sigma, \quad \langle B_k \alpha, \beta \rangle = \sum_{\sigma \in S_k} a_\sigma(\alpha, \beta).$$

Every U_σ fixes $x^{\otimes k}$ for every $x \in E^\circ$. Consequently the last scalar is independent of the unit letter used to define V_k . The same slice identity, now tested against arbitrary endpoint and tail vectors and arbitrary β , gives the first equality in (20). These tests determine the whole output on level k , and the other levels have already been excluded. This also proves the assertion for every endpoint $d_0 \in E$, with no boundedness requirement on its representative. No self-adjointness of x is used: the relevant cross contraction is $\tau(xz) = \langle x, z^* \rangle$.

The same argument applies to G^* . Its induced operator on J_k is $V_k^* G^* V_k = (V_k^* G V_k)^* = B_k^*$, proving the second equality. $\square$

At level zero the lemma says $G|_{\mathcal{H}_0} = I_E \otimes B_0$. Since $B = L^2(A_0)$ is the subspace $\mathbb{C}1 \otimes B$ there, the quantity to be estimated in Theorem 3.1, after symmetrization, is exactly

$$\|P_B[G, L_s]|_B\| = \|[B_0, L_s|_B]\|.$$

4 Propagation from small corners

Continue with the notation and hypotheses of Theorem 3.1, and set $H = L^2(\mathcal{L})$. After Lemma 3.3, we may use the symmetrized operator G and the operators $B_k \in \mathcal{B}(J_k)$ from Lemma 3.5. Our goal is to compare the action of B_k with that of B_0 along repeated letters, with an error independent of k . We first extract a consequence of the small-corner hypothesis that needs only the exact commutation with L_D .

Lemma 4.1 (Full columns from a small corner). *For every projection $r \in \mathcal{L}$ with $\tau(r) \leq t$, there is $b \in \mathcal{L}$ with $b = rb$ such that*

$$\|(G - R_b)|_{Hr}\| \leq c_0 \epsilon. \quad (21)$$

The same assertion holds for G^ .*

Proof. The case $r = 0$ is immediate. Fix $r \neq 0$ and a projection $e \in D$ with $\tau(e) = t$. The factoriality established in Lemma 3.2 and comparison of projections give $r' \leq e$ and a partial isometry $v \in \mathcal{L}$ such that $v^*v = r$ and $vv^* = r'$. Right multiplication by v^* is a unitary left $e\mathcal{L}e$ -module map

$$eHr \longrightarrow eHr', \quad \zeta \longmapsto \zeta v^*.$$

Indeed its inverse is right multiplication by v , and

$$\|\zeta v^*\|_2^2 = \tau(v\zeta^*\zeta v^*) = \tau(\zeta^*\zeta r) = \|\zeta\|_2^2.$$

These equalities also hold for L^2 vectors by continuity. The vector r' is cyclic for the left action of $e\mathcal{L}e$ on eHr' , since $(e\mathcal{L}e)r' = e\mathcal{L}r'$ is dense in that Hilbert space. Thus eHr is a cyclic domain representation of the unital algebra $e\mathcal{L}e$.

Since G commutes with L_e , its restriction defines a bounded map $G_e : eHr \rightarrow eH$. On these domain and codomain representations the map

$$\Delta(a) = G_e L_a|_{eHr} - L_a G_e, \quad a \in e\mathcal{L}e,$$

is a rectangular derivation of norm at most ϵ , by (16). Lemma 2.4 therefore supplies an exact intertwiner $T : eHr \rightarrow eH$ with

$$T L_a|_{eHr} = L_a T \quad (a \in e\mathcal{L}e), \quad \|G_e - T\| \leq c_0 \epsilon.$$

The value $t = 1/N$ allows an extension that preserves this norm. Choose $v_1, \dots, v_N \in D$ with

$$v_i^* v_i = e, \quad v_i v_i^* \perp v_j v_j^* \quad (i \neq j), \quad \sum_{i=1}^N v_i v_i^* = 1.$$

The maps $(\xi_i) \mapsto \sum_i v_i \xi_i$ identify $(eHr)^N$ with Hr and $(eH)^N$ with H unitarily. Define

$$\tilde{T} \left(\sum_{i=1}^N v_i \xi_i \right) = \sum_{i=1}^N v_i T \xi_i, \quad \xi_i \in eHr.$$

In these coordinates $\tilde{T}$ is $T^{(N)}$, and $G|_{Hr}$ is $G_e^{(N)}$, because G commutes with each L_{v_i} . Consequently

$$\|G|_{Hr} - \tilde{T}\| = \|G_e - T\| \leq c_0 \epsilon.$$

For $a \in \mathcal{L}$, its left action has matrix entries $L_{v_i^* a v_j}$ in both coordinate systems. Every $v_i^* a v_j$ lies in $e\mathcal{L}e$, so the intertwining identity for T implies $\tilde{T} L_a|_{Hr} = L_a \tilde{T}$.

Extend $\tilde{T}$ by zero on $H(1-r)$. Since R_r commutes with all left multiplications, this extension belongs to $L(\mathcal{L})'$ and hence is R_b for a bounded $b \in \mathcal{L}$, by the tracial commutation theorem. Its vanishing on $H(1-r)$ says $R_b = R_b R_r$, which, with the convention $R_b \xi = \xi b$, gives $b = rb$. This proves (21). Finally G^* commutes with L_D and $[G^*, L_a] = -[G, L_a^*]^*$, so all the same hypotheses and estimates hold for G^* . $\square$

Lemma 4.2 (Propagation along repeated letters). *For every $k \geq 1$ and $v \in B = L^2(A_0)$,*

$$\|B_k(s^{\otimes k} \otimes v) - s^{\otimes k} \otimes B_0 v\| \leq 2c_0 \epsilon \|v\|_2. \quad (22)$$

This also holds with B_k^ and B_0^* in place of B_k and B_0 .*

Proof. Fix $k \geq 1$ and first take $v \in A_0$; the zero vector is harmless. Choose a nonzero projection $q \in D$ with $a = \tau(q) \leq t$. The left support of qv is dominated by q . Its right support r is equivalent to its left support by polar decomposition, so $\tau(r) \leq a \leq t$. Choose $b = rb$ as in Lemma 4.1 for this r , and put

$$y = qvb - qB_0 v.$$

The level-zero identity in (20) gives $G(qv) = qB_0 v$, and the tensor decomposition gives $\|qv\|_2 = \sqrt{a} \|v\|_2$. It follows that

$$qy = y, \quad \|y\|_2 \leq c_0 \epsilon \sqrt{a} \|v\|_2. \quad (23)$$

The vector $B_0 v$ need only belong to $L^2(A_0)$; every expression here uses bounded left multiplication on this vector.

Let z be a Haar unitary of qDq for its normalized trace, and set $x_n = a^{-1/2}z^n$ for $n \geq 1$. Then

$$\tau(x_n) = 0, \quad \|x_n\|_2 = 1, \quad x_n^* x_n = x_n x_n^* = a^{-1}q, \quad x_n, x_n^* \rightharpoonup 0 \text{ in } L^2(D).$$

The weak convergence follows, for example, from orthonormality of the positive and negative powers. The reduced words $\zeta_n = (sx_n)^k v$ have norm $\|v\|_2$ and right support at most r , because their last x_n satisfies $x_n q = x_n$. With $A_n = (sx_n)^{k-1}s$, we have

$$\zeta_n b - (sx_n)^k B_0 v = A_n x_n y.$$

Thus (21) yields

$$\|G\zeta_n - (sx_n)^k B_0 v\|_2 \leq c_0 \epsilon \|v\|_2 + \|A_n x_n y\|_2. \quad (24)$$

To estimate the second term, write (18) as $H = E \otimes F$, where

$$F = \bigoplus_{l \geq 0} (E^\circ)^{\otimes l} \otimes J_l,$$

and let P_{sc} be projection onto $\mathbb{C}1 \otimes F$. This projection keeps every word level and selects the scalar component of its first D factor. On this decomposition left multiplication by x_n acts on E . With inner products linear in the first variable, define $\ell_n(\eta) = \langle \eta, x_n^* \rangle$. Then

$$P_{\text{sc}}(x_n y) = 1 \otimes (\ell_n \otimes I_F)y, \quad \|\ell_n\| = 1.$$

For each elementary tensor this expression tends to zero in norm, since $x_n^* \rightharpoonup 0$. Approximating the fixed vector y by finite tensor sums, and using the common bound $\|\ell_n\| = 1$, proves

$$\|P_{\text{sc}}(x_n y)\|_2 \longrightarrow 0. \quad (25)$$

On $(1 - P_{\text{sc}})H$, multiplication by A_n prepends a reduced centered word ending in s to a leading centered D letter. Each prefix letter has L^2 norm one, so this restriction is an isometry. On $P_{\text{sc}}H$ we use the bound $\|L_{A_n}\| \leq a^{-(k-1)/2}$. Also $qy = y$ gives

$$\|x_n y\|_2^2 = \langle L_{x_n^* x_n} y, y \rangle = a^{-1} \|y\|_2^2.$$

Consequently

$$\|A_n x_n y\|_2 \leq \|(1 - P_{\text{sc}})x_n y\|_2 + a^{-(k-1)/2} \|P_{\text{sc}} x_n y\|_2,$$

and (25) implies

$$\limsup_n \|A_n x_n y\|_2 \leq a^{-1/2} \|y\|_2 \leq c_0 \epsilon \|v\|_2.$$

Here q, k, v, b, y are all fixed before n tends to infinity. The possibly large prefix bound therefore disappears with the vanishing coefficient; a convergence rate uniform in a or k is unnecessary.

By the full-output identity (20), the left side of (24) equals

$$\|B_k(s^{\otimes k} \otimes v) - s^{\otimes k} \otimes B_0 v\|$$

for every n : its endpoint and repeated D factors all have norm one. Taking the limit superior proves (22) for bounded v . Density of A_0 in B and boundedness of the operators extend it to all $v \in B$. Applying the same argument to G^* , using the adjoint identities of Lemma 3.5, gives the final assertion. $\square$

The propagation estimate now controls all word levels with the same error. We assemble orthogonal copies of B from these levels; this will allow a single compressed commutator to test B_0 .

Choose $e \in D$ with $\tau(e) = t$ and $p \in D$ with $e \leq p$ and $\tau(p) = 1/2$; this is possible since $t \leq 1/2$. Set

$$d = 2p - 1, \quad w = ds, \quad u = \tau_p(e) = 2t, \quad \tau_p = 2\tau|_{p\mathcal{L}p}. \quad (26)$$

The elements d and s are free centered symmetries. For $k \geq 0$ define

$$U_k : B \longrightarrow eH, \quad U_k v = t^{-1/2} e w^{-k} v = t^{-1/2} e (sd)^k v.$$

In (18), this is the tensor with endpoint $t^{-1/2}e$, repeated D letter d , and last factor $s^{\otimes k} \otimes v$. Hence each U_k is an isometry, its range belongs to level k , and these ranges are mutually orthogonal. Put

$$\mathcal{K} = \overline{\bigoplus_{k \geq 0} U_k B} \subset eH.$$

Via the U_k , identify $\mathcal{K}$ with $\ell^2(\mathbb{N}_0) \otimes B$, where $\mathbb{N}_0 = \{0, 1, \dots\}$, and set $J = I \otimes B_0$. Denote the inclusion $\mathcal{K} \hookrightarrow H$ by ι .

Lemma 4.3 (Compression of the commutator). *For every $h \in e\mathcal{L}e$,*

$$\|[J, \iota^* L_h \iota]\| \leq (1 + 4c_0)\epsilon \|h\|. \quad (27)$$

Proof. Lemma 4.2 and (20) give

$$\|GU_k v - U_k B_0 v\|_2 \leq 2c_0 \epsilon \|v\|_2,$$

with zero error for $k = 0$. Each error lies in level k by Lemma 3.5. Their squared norms therefore add on finite orthogonal sums. Passing to the closure, and repeating the argument for G^* , gives the full-codomain estimates

$$\|G\iota - \iota J\| \leq 2c_0 \epsilon, \quad \|G^* \iota - \iota J^*\| \leq 2c_0 \epsilon. \quad (28)$$

Write $R = G\iota - \iota J$, $R' = G^* \iota - \iota J^*$, and $T_h = \iota^* L_h \iota$. Taking the adjoint of the second error identity gives $\iota^* G = J\iota^* + (R')^*$. A direct expansion now yields

$$\iota^* [G, L_h] \iota = [J, T_h] + (R')^* L_h \iota - \iota^* L_h R.$$

Apply (16) to the first term and (28) to the other two. Their bounds are $\epsilon \|h\|$, $2c_0 \epsilon \|h\|$, and $2c_0 \epsilon \|h\|$, proving (27). $\square$

5 Free compression and the Hankel test

Retain e, p, d, w, u , the isometries U_k , and the subspace $\mathcal{K}$ from the preceding section. We construct $h \in e\mathcal{L}e$ whose norm is of order u and whose compression to $\mathcal{K}$ has a Hankel component of the same order, with a uniform lower bound after division by u .

5.1 The dihedral corner

Let $\mathcal{C} = W^*(d, s)$. Freeness of the centered symmetries d, s shows that their alternating reduced words, together with 1, form an orthonormal word basis of $L^2(\mathcal{C})$. These are precisely the group words of the infinite dihedral group. In particular,

$$w = ds \text{ is a Haar unitary,} \quad dwd = w^{-1}, \quad p = (1 + d)/2.$$

Write m for Haar probability measure on $\mathbb{T}$.

Lemma 5.1 (The abelian corner and corner freeness). *The algebra $p\mathcal{C}p$ is identified with the even functions on the circle by*

$$\begin{aligned} p\mathcal{C}p &= \{f(w)p : f \in L^\infty(\mathbb{T}, m), f(z) = f(z^{-1})\}, \\ \tau_p(f(w)p) &= \int_{\mathbb{T}} f dm. \end{aligned} \tag{29}$$

The map $f \mapsto f(w)p$ in this formula is a faithful normal unital $*$ -isomorphism onto the corner. Moreover, pDp and $p\mathcal{C}p$ are free with respect to the normalized trace τ_p .

Proof. Every group word has the form w^l or $w^l d$, $l \in \mathbb{Z}$. Since $dp = p$ and $d w^l d = w^{-l}$,

$$p w^l d p = p w^l p = \frac{w^l + w^{-l}}{2} p.$$

An even function of w commutes with d and hence with p . Thus $f \mapsto f(w)p$ is a normal $*$ -homomorphism on the even-function algebra, with unit p . The group trace gives

$$\tau_p(f(w)p) = 2\tau(f(w)p) = \tau(f(w)) + \tau(f(w)d) = \int_{\mathbb{T}} f dm,$$

initially for trigonometric polynomials and then for bounded functions by L^2 approximation. In particular the map is faithful, and its range is a von Neumann algebra. The compressed group-word span is strongly dense in $p\mathcal{C}p$ and consists of even Laurent polynomials times p . This proves (29) and the isomorphism assertion. It also gives

$$\|f(w)p\| = \|f\|_\infty, \quad \|f(w)p\|_{2, \tau_p} = \|f\|_{L^2(m)}.$$

For freeness, let $a_1, \dots, a_j \in pDp$ be centered for τ_p . Then $\tau(a_i) = 0$ and $a_i d = a_i$. The centered trigonometric span in $p\mathcal{C}p$ is generated by the elements $p w^l p$, $l \geq 1$. For $l_1, \dots, l_j \geq 1$, absorbing the corner projections and using traciality gives

$$\begin{aligned} \tau_p(a_1 p w^{l_1} p \cdots a_j p w^{l_j} p) &= 2\tau(a_1 w^{l_1} \cdots a_j w^{l_j}) \\ &= 2\tau(a_1 s (ds)^{l_1-1} \cdots a_j s (ds)^{l_j-1}) = 0. \end{aligned}$$

All the displayed D letters are centered, and s is centered and free from the whole algebra D , by $\mathcal{L} = D * A_0$. Thus the last word is a reduced alternating word of trace zero. Fejér means of an even bounded zero-mean function are even zero-mean trigonometric polynomials, are uniformly bounded, and converge in $L^2(m)$. Approximating each corner letter in this manner extends the vanishing to all cyclic alternating products of even length. Products of odd alternating length have their endpoints in the same algebra. Move one endpoint cyclically next to the other, split their product into its centered and scalar parts, and apply the even-length case and induction on the length. The length-one case is centeredness. This proves freeness. $\square$

5.2 The expectation and the norm of a free compression

Lemma 5.2 (Free compression). *For $b = b^* \in p\mathcal{C}p$ with $\tau_p(b) = 0$,*

$$E_C(ebe) = u^2b, \quad \|ebe\| \leq 2\sqrt{u} \|b\|_{2,\tau_p} + u\|b\|, \quad u = \tau_p(e). \quad (30)$$

Here E_C is the trace-preserving conditional expectation from $\mathcal{L}$ onto $\mathcal{C}$.

Proof. Put $\mathcal{A} = pDp$, $\mathcal{Z} = p\mathcal{C}p$, and $a = e - up \in \mathcal{A}$, so $\tau_p(a) = 0$. Since $p \in \mathcal{C}$, bimodularity shows that $E_C(ebe)$ belongs to $\mathcal{Z}$. On $p\mathcal{L}p$, its restriction is the trace-preserving expectation onto $\mathcal{Z}$ for τ_p . Expand

$$ebe = u^2b + u(ab + ba) + aba.$$

Freeness gives zero expectation for ab and ba . For the last term, test against any $c \in \mathcal{Z}$ and write $c = c^\circ + \tau_p(c)p$. Freeness and traciality give

$$\tau_p(c^\circ aba) = 0, \quad \tau_p(aba) = \tau_p(ba^2) = \tau_p(b)\tau_p(a^2) = 0.$$

Thus $E_{\mathcal{Z}}(aba) = 0$ also. This proves the expectation formula.

For the norm estimate work in

$$H_c = L^2(W^*(\mathcal{A}, \mathcal{Z}), \tau_p).$$

This representation is faithful, so its operator norm is the algebra norm. The free-word decomposition, grouping a possible initial $\mathcal{A}$ letter, takes the form

$$H_c = L^2(\mathcal{A}, \tau_p) \otimes F_0, \quad F_0 = \mathbb{C}\Omega \oplus F_0^\circ,$$

where F_0° is the closed span of reduced words starting in the centered L^2 subspace $\mathcal{Z}^\circ$. Let Q_0 be the projection of F_0 onto the complement of the vacuum, and let Π project in H_c onto words starting in $\mathcal{Z}^\circ$. Then

$$\Pi = |1_{\mathcal{A}}\rangle\langle 1_{\mathcal{A}}| \otimes Q_0, \quad L_e = \lambda_{\mathcal{A}}(e) \otimes I_{F_0},$$

where $1_{\mathcal{A}} = p$ has $L^2(\tau_p)$ norm one. Consequently

$$\|L_e\Pi\| = \|\Pi L_e\| \leq \|e\|_{2,\tau_p} = \sqrt{u}. \quad (31)$$

The following three blocks are the standard creation, annihilation, and diagonal decomposition of a centered free-product letter; see Ricard and Xu [26, Section 1, Lemmas 2.1–2.3 and Corollary 2.4]. We give the bounds needed here directly. Let $T = L_b$ on H_c . Because b is centered, on the vacuum and on words starting in $\mathcal{A}^\circ$ its left action simply prepends the letter b . Therefore

$$(1 - \Pi)T(1 - \Pi) = 0, \quad \|\Pi T(1 - \Pi)\| \leq \|b\|_{2,\tau_p}.$$

Self-adjointness gives the same bound for $(1 - \Pi)T\Pi$, and $\|\Pi T\Pi\| \leq \|b\|$. Decomposing T into these three blocks, using $\|L_e\| \leq 1$, and applying (31), we obtain

$$\begin{aligned} \|ebe\| &= \|L_e T L_e\| \leq \|L_e \Pi T(1 - \Pi) L_e\| + \|L_e(1 - \Pi) T \Pi L_e\| + \|L_e \Pi T \Pi L_e\| \\ &\leq \sqrt{u} \|b\|_{2,\tau_p} + \sqrt{u} \|b\|_{2,\tau_p} + u\|b\|. \end{aligned}$$

This proves (30). □

5.3 Fourier coefficients of the compressed operator

The preceding lemma gives both the expectation and the upper norm bound for a corner compression. We now compute its matrix entries on $\mathcal{K}$; their dependence on the sum of the indices will isolate the commutator with L_s .

Let $f \in L^\infty(\mathbb{T}, m)$ be real, even, and of mean zero, and set

$$b_f = f(w)p, \quad h = eb_fe, \quad \gamma_l = \int_{\mathbb{T}} z^l f(z) dm(z) \quad (l \in \mathbb{Z}).$$

Then b_f meets the hypotheses of Lemma 5.2. We claim that, for $j, k \geq 0$,

$$\begin{aligned} U_j^* L_h U_k &= L_{t^{-1}E_{A_0}(w^j h w^{-k})}|_B \\ &= u(\gamma_{j-k} I_B + \gamma_{j+k+1} L_s)|_B. \end{aligned} \tag{32}$$

For the first equality, test on $v, z \in A_0$. Since $eh = he = h$,

$$\begin{aligned} \langle L_h U_k v, U_j z \rangle &= t^{-1} \tau(z^* w^j h w^{-k} v) \\ &= \left\langle t^{-1} E_{A_0}(w^j h w^{-k}) v, z \right\rangle. \end{aligned}$$

Density then gives the operator identity on B .

The reduced-word decomposition shows that on $W^*(D, s)$ the expectation E_{A_0} equals the expectation onto $W^*(s) = \mathbb{C}1 + \mathbb{C}s$: every reduced word containing a centered D letter is orthogonal to $L^2(A_0)$. Because $W^*(s) \subset \mathcal{C}$, this latter expectation factors through $E_{\mathcal{C}}$. By (30) and bimodularity,

$$E_{\mathcal{C}}(w^j h w^{-k}) = u^2 w^j b_f w^{-k}.$$

Using $dw^{-k} = w^k d$ and $d = ws$, compute

$$w^j p w^{-k} = \frac{w^{j-k} + w^{j+k} d}{2} = \frac{w^{j-k} + w^{j+k+1} s}{2}.$$

The group trace implies

$$E_{W^*(s)}(f(w)w^l) = \gamma_l 1, \quad E_{W^*(s)}(f(w)w^l s) = \gamma_l s.$$

For instance this follows on Laurent polynomials because only the identity power contributes, and then by L^2 approximation. Hence

$$t^{-1} E_{A_0}(w^j h w^{-k}) = \frac{u^2}{2t} (\gamma_{j-k} 1 + \gamma_{j+k+1} s) = u(\gamma_{j-k} 1 + \gamma_{j+k+1} s),$$

which proves (32), including its normalization and its Hankel index.

The matrices

$$T_f = (\gamma_{j-k})_{j,k \geq 0}, \quad H_f = (\gamma_{j+k+1})_{j,k \geq 0}$$

define bounded operators on $\ell^2(\mathbb{N}_0)$ of norm at most $\|f\|_\infty$. To see this directly, decompose

$$L^2(\mathbb{T}, m) = H_- \oplus H_+, \quad H_- = \overline{\text{span}}\{z^{-j} : j \geq 0\}, \quad H_+ = \overline{\text{span}}\{z^{k+1} : k \geq 0\}.$$

In the indicated orthonormal bases, T_f and H_f are respectively the H_- , H_- and H_- , H_+ blocks of multiplication by f . Equation (32), first on finite coordinate vectors and then by continuity, therefore gives

$$\iota^* L_h \iota = u(T_f \otimes I_B + H_f \otimes L_s)|_B. \tag{33}$$

As $J = I \otimes B_0$, it follows that

$$\| [J, \iota^* L_h \iota] \| = u \| H_f \| \| [B_0, L_s|_B] \|. \quad (34)$$

Here we used the operator norm identity $\|A \otimes B\| = \|A\| \|B\|$ for Hilbert-space tensor products.

Completion of the proof of Theorem 3.1. Put $\theta = u/4 = 1/(2N)$, so $0 < \theta \leq 1/4$, and use angular coordinates $z = e^{i\alpha}$, $-\pi < \alpha \leq \pi$, to define

$$f(e^{i\alpha}) = \mathbf{1}_{[-\theta, \theta]}(\alpha) - \frac{\theta}{\pi}.$$

This function is real, even, and of mean zero. The trace formula in (29) gives

$$\|b_f\| \leq 1, \quad \|b_f\|_{2, \tau_p}^2 = \frac{\theta}{\pi} \left(1 - \frac{\theta}{\pi}\right) \leq u.$$

Consequently (30) gives

$$\|h\| = \|eb_f e\| \leq 3u. \quad (35)$$

For every $l \geq 1$ the Fourier coefficient is

$$\gamma_l = \frac{\sin(l\theta)}{\pi l}.$$

If $0 \leq j, k < N$, then $1 \leq j + k + 1 \leq 2N - 1$ and $(j + k + 1)\theta < 1 < \pi/2$. The inequality $\sin x \geq 2x/\pi$ on $[0, \pi/2]$ therefore gives

$$(H_f)_{jk} \geq \frac{2\theta}{\pi^2}.$$

Testing the leading N by N block on its constant unit vector, and using $\theta N = 1/2$, proves the uniform bound

$$\|H_f\| \geq \frac{2\theta N}{\pi^2} \geq \frac{1}{2\pi^2}. \quad (36)$$

Combine (34), (27), (35), and (36) to obtain

$$\frac{u}{2\pi^2} \| [B_0, L_s|_B] \| \leq 3u(1 + 4c_0)\epsilon.$$

Cancelling $u > 0$ yields

$$\| [B_0, L_s|_B] \| \leq 6\pi^2(1 + 4c_0)\epsilon = c_1\epsilon.$$

The level-zero case of (20) identifies $G|_B$ with B_0 , so this is exactly the required compression estimate for the symmetrized operator. Lemma 3.3 preserves the compression of $[G, L_s]$ between vectors of B . The same estimate therefore holds for the original operator in Theorem 3.1, completing its proof. $\square$

6 Transfer to finite factors

We now transfer Theorem 3.1 to matrix commutators in finite sums of the standard representation of a separable-predual II_1 factor. All traces on matrix amplifications are normalized.

We first average the operator over an irreducible subfactor generated by an increasing sequence of full matrix algebras. After matrix amplification, the ultrapower of the amplified subfactor contains another subfactor generated by increasing full matrix algebras and free from the constant copy of the amplified ambient factor. Extending and compressing the averaged operator then puts it in the setting of Theorem 3.1. Compression to constant vectors transfers the estimate back to the matrix test, with the fixed error from the initial averaging.

The existence of an irreducible hyperfinite subfactor in a separable-predual II_1 factor is due to Popa [23]; we include a direct proof of the increasing-matrix form used below.

Lemma 6.1 (An irreducible increasing-matrix subfactor). *Every separable-predual II_1 factor M contains a unital subfactor R with $R' \cap M = \mathbb{C}1$, generated by an increasing sequence of unital full matrix algebras whose sizes tend to infinity.*

Proof. We first give an approximation step in an arbitrary II_1 factor N . If $y = y^* \in N$ and $\eta > 0$, there is a unital full matrix algebra $F \subset N$, of size at least two, such that

$$\|E_{F' \cap N}(y) - \tau(y)1\|_2 < \eta.$$

Indeed, approximate y in L^2 by a spectral step $y_0 = \sum_{i=1}^k \lambda_i p_i$, where the p_i partition the identity. For a large integer b , split off $\lfloor b\tau(p_i) \rfloor$ subprojections of trace $1/b$ in each p_i . The remaining projection has trace

$$\frac{b - \sum_{i=1}^k \lfloor b\tau(p_i) \rfloor}{b} < \frac{k}{b}.$$

Its numerator is an integer, so it too splits into projections of trace $1/b$. These pieces form a partition of the identity by b equivalent projections and hence extend to matrix units for a unital $F \cong M_b(\mathbb{C})$. Define $y_1 \in F$ to have value λ_i on the pieces taken from p_i , and value zero on the remaining pieces. Then

$$\|y - y_1\|_2 \leq \|y - y_0\|_2 + \|y_0\| \sqrt{k/b}.$$

The expectation of y_1 onto $F' \cap N$ is $\tau(y_1)1$. Contractivity of this expectation on L^2 , and $|\tau(y - y_1)| \leq \|y - y_1\|_2$, give the required estimate by choosing the spectral approximation and then b .

Choose a countable L^2 -dense subset $\{x_j : j \geq 1\}$ of the self-adjoint unit ball of M , and enumerate all requirements $(j, k) \in \mathbb{N}^2$. Begin with $F_0 = \mathbb{C}1$. If the current stage is a unital full matrix algebra F_n , then $N_n = F'_n \cap M$ is a II_1 factor: matrix units identify M with a matrix algebra over a corner and N_n with that corner. For the next requirement (j, k) , apply the approximation step in N_n to $y = E_{N_n}(x_j)$, with error less than $1/k$, obtaining $F \subset N_n$. Set $F_{n+1} = F_n \vee F$. This is again a full matrix algebra, its size is at least twice that of F_n , and $N_{n+1} = F' \cap N_n$. The identity $E_{N_{n+1}} = E_{N_{n+1}} E_{N_n}$ gives

$$\|E_{N_{n+1}}(x_j) - \tau(x_j)1\|_2 < 1/k.$$

All previously obtained bounds persist because the relative commutants decrease and their trace expectations compose.

Let $R = (\bigcup_n F_n)''$ and $N = R' \cap M = \bigcap_n N_n$. Since $E_N = E_N E_{N_n}$, every requirement remains valid with N in place of N_{n+1} . Thus $E_N(x_j) = \tau(x_j)1$ for every j . Density gives $E_N(x) = \tau(x)1$ for all $x \in M$, so $N = \mathbb{C}1$. In particular R is a factor. It is finite and contains full matrix algebras of unbounded size, hence is of type II_1 , with the asserted increasing-matrix presentation. $\square$

For a finite tracial von Neumann algebra P and a nonprincipal ultrafilter ω on $\mathbb{N}$, write $P^\omega = \ell^\infty(\mathbb{N}, P)/\mathcal{I}_\omega$, where $\mathcal{I}_\omega$ consists of the uniformly bounded sequences (a_i) with $\lim_\omega \|a_i\|_2 = 0$. Its trace is $\tau_\omega([a_i]) = \lim_\omega \tau(a_i)$, and P embeds by constant sequences. The lemma below places a free subfactor in such an ultrapower while retaining its inclusion in the prescribed subalgebra Q^ω .

Lemma 6.2 (Free position in the full ultrapower). *Let $Q \subset P$ be a unital irreducible inclusion of II_1 factors, with P of separable predual. For any nonprincipal ultrafilter ω on $\mathbb{N}$, there is a subfactor $D \subset Q^\omega$, generated by increasing unital full matrix algebras of unbounded sizes, which is free from the constant copy of P in the tracial ultrapower P^ω .*

Proof. We use the ultraproduct-of-subalgebras case of Popa's independence theorem [25, Theorem 0.1(a); see also Theorem 4.3]. For von Neumann subalgebras $A, B \subset M$, the notation $A \prec_M B$ means Popa's intertwining of a nonzero corner of A into a finite matrix amplification of B . Its hypotheses are finite factors M_i with $\dim M_i \rightarrow \infty$, and unital von Neumann subalgebras $Q_i \subset M_i$ satisfying $Q_i \not\prec_{M_i} (Q'_i \cap M_i)$. Write $\mathbf{M} = \prod_\omega M_i$, $\mathbf{Q} = \prod_\omega Q_i$, and $C = \mathbf{Q}' \cap \mathbf{M}$. Given an L^2 -separable subspace $X \subset \mathbf{M} \ominus C$, the theorem supplies a diffuse abelian von Neumann algebra $A \subset \mathbf{Q}$ such that

$$E_C(x_0 a_1 x_1 \cdots a_n x_n) = 0$$

whenever $n \geq 1$, $x_0 \in X \cup \{1\}$, $x_i \in X$ for $1 \leq i \leq n$, and $a_i \in A \ominus \mathbb{C}1$. In particular, if $C = \mathbb{C}1$ and $T \subset \mathbf{M}$ is a separable-predual von Neumann subalgebra, take $X = T \ominus \mathbb{C}1$. The moment identities, traciality, and decomposition into scalar and centered parts show that A is free from T . Here separability is for L^2 , as stipulated in Popa [25, Section 1.1].

We check these hypotheses for the constant sequences $M_i = P$ and $Q_i = Q$. The dimension condition holds because these factors are infinite-dimensional. If $Q \prec_P \mathbb{C}1$, the intertwining data would include a nonzero normal $*$ -homomorphism from a nonzero corner of Q into a finite-dimensional matrix algebra. Such a homomorphism from a factor is faithful, whereas a II_1 factor has arbitrarily large families of nonzero orthogonal projections. This is impossible. Since $Q' \cap P = \mathbb{C}1$, the non-intertwining hypothesis follows.

We also verify the full ultrapower relative commutant:

$$(Q^\omega)' \cap P^\omega = \mathbb{C}1.$$

For a uniformly bounded representative (x_i) of an element of this commutant, commuting with all sequences of unitaries from Q implies

$$d_i := \sup_{v \in \mathcal{U}(Q)} \|vx_i v^* - x_i\|_2 \rightarrow_\omega 0.$$

Otherwise one could choose a witnessing unitary at each coordinate on an ω -large set. The L^2 -closed convex hull of the Q -unitary orbit of x_i has a unique vector of minimum norm. It is Q -central, has trace $\tau(x_i)$, and belongs to P : the hull stays in the operator-norm ball of radius $\|x_i\|$, which is L^2 -closed by weak- $*$ compactness and trace pairings. Irreducibility makes this vector $\tau(x_i)1$. Every point in the hull is within d_i of x_i , so $\|x_i - \tau(x_i)1\|_2 \leq d_i$. Bounded scalar sequences equal their scalar ultralimit in the tracial ultrapower. This proves the claim, which concerns all of Q^ω , rather than merely the constant copy of Q .

Choose a Haar unitary u_1 in the diffuse abelian algebra obtained by applying Popa's theorem to the constant P . Then $u_1 \in Q^\omega$ is free from P . The algebra $W^*(P, u_1)$ has separable L^2 space: countably many words in generators of P , in u_1 , and in their adjoints have linear span dense in L^2 . A second application of the same theorem therefore gives a Haar unitary $u_2 \in Q^\omega$

free from $W^*(P, u_1)$. Expanding centered reduced words shows that $P, W^*(u_1), W^*(u_2)$ are jointly free. Thus $W^*(u_1, u_2)$ is free from P and, by its free Haar moments, is trace-isomorphic to $L(\mathbb{F}_2)$. This is a separable-predual II_1 factor: the nontrivial conjugacy classes of $\mathbb{F}_2$ are infinite, so a central square-summable Fourier coefficient function is supported at the identity. Apply Lemma 6.1 inside $W^*(u_1, u_2)$ to obtain D . Freeness passes to its subalgebra D , proving the lemma. The ambient tracial ultrapower throughout is P^ω . $\square$

Proposition 6.3 (Finite standard multiplicities). *Let M be a separable-predual II_1 factor, and let M act on $H_r = L^2(M)^r$ by common left multiplication, where $r \geq 1$ is finite. For every $Y \in \mathcal{B}(H_r)$, every finite $h \geq 1$, and every $X \in M_h(M)$,*

$$\|[Y^{(h)}, X]\| \leq (3c_1 + 2) g_M(Y) \|X\|.$$

Here X denotes its entrywise action on H_r^h ; the constant is independent of M, r, h .

Proof. Put $\delta = g_M(Y)$. By homogeneity it suffices to consider $\|X\| \leq 1$. We first arrange a symmetry test on a square array of standard copies. Choose $h_0 \geq h$ so that $m = 2h_0 \geq r$, extend Y by zero to H_m , and retain the notation Y . The copy projection reduces the common action of M , so this extension still has scalar commutator bound δ . For $X' = X \oplus 0 \in M_{h_0}(M)$, define in $P = M_m(M)$

$$s = \begin{pmatrix} (1 - X'X'^*)^{1/2} & X' \\ X'^* & -(1 - X'^*X')^{1/2} \end{pmatrix}.$$

Functional calculus gives $(1 - X'X'^*)^{1/2}X' = X'(1 - X'^*X')^{1/2}$, and hence $s = s^* = s^{-1}$. Traciality gives equal moments for $X'X'^*$ and X'^*X' , including the zeroth moment; polynomial approximation on $[0, 1]$ gives equal traces for their displayed square roots. Consequently $\tau_P(s) = 0$.

Write vectors of H_m^m as arrays (ξ_{ij}) , with i the outer amplification coordinate and j the copy coordinate. The map $(\xi_{ij}) \mapsto \sqrt{m}(\xi_{ij})$ is a unitary from H_m^m to $L^2(P)$, since $\|(\xi_{ij})\|_2^2 = m^{-1} \sum_{i,j} \|\xi_{ij}\|_2^2$ on the latter space. Under this identification, the matrix action of P is left multiplication. The upper-right block of $[Y^{(m)}, L_s]$ is the commutator with X' . Restricting its row coordinates to the original h and its copy coordinates to the original r recovers $[Y^{(h)}, X]$ for the original operator. It is therefore enough to bound $[Y^{(m)}, L_s]$.

Choose $R \subset M$ from Lemma 6.1, with matrix stages F_n . On H_m , average Y over the compact unitary group of each F_n , using common left multiplication, and take a weak operator cluster point Y_0 as n tends to infinity. For every such unitary v , $\|L_v Y L_v^* - Y\| \leq \delta$. The averages, and hence their cluster point, are within δ of Y . An average over a later stage commutes with every earlier stage, so

$$[Y_0, L_R] = 0, \quad \|Y_0 - Y\| \leq \delta, \quad g_M(Y_0) \leq 3\delta.$$

The first identity extends from the matrix stages to R by strong density; the norm bounds pass to the cluster point by weak operator closedness of norm balls.

On $L^2(P) = H_m^m$, set $S = Y_0^{(m)}$ and $Q = M_m(R)$. Matrix units and $R' \cap M = \mathbb{C}1$ imply $Q' \cap P = \mathbb{C}1$. Moreover S commutes with L_Q , since it has the same operator Y_0 on every row and Y_0 commutes with every entry from R . For $e_0 = e_{11} \otimes 1$, the space $e_0 L^2(P)$ is one row of m standard copies, with $e_0 P e_0 \cong M$ acting by common left multiplication. Since S commutes with L_{e_0} , it follows on the full Hilbert space that

$$\|[S, L_z]\| \leq 3\delta \|z\| \quad (z \in e_0 P e_0).$$

If $f \in Q$ is a projection with $\tau_P(f) \leq 1/m$, projection comparison in the factor Q gives a unitary $v \in Q$ with $f \leq ve_0v^*$. Conjugating by L_v , which commutes with S , yields

$$\|[S, L_z]\| \leq 3\delta\|z\| \quad (z \in fPf, f \in Q, \tau_P(f) \leq 1/m).$$

We have obtained the same small-corner estimate for every projection of Q of trace at most $1/m$. The next step transports that estimate to the ultrapower, where Lemma 6.2 supplies the free subfactor. Fix a nonprincipal ultrafilter ω on $\mathbb{N}$. Recall the tracial quotient and distinguish it from the Hilbert ultrapower:

$$P^\omega = \ell^\infty(\mathbb{N}, P) / \mathcal{I}_\omega, \quad \mathcal{I}_\omega = \{(a_i) : \lim_\omega \|a_i\|_2 = 0\},$$

$$\mathcal{H} = (L^2(P))^\omega_{\text{Hilb}}.$$

The Hilbert ultrapower uses bounded Hilbert-space sequences modulo those whose Hilbert norms tend to zero along ω . Sending the class of a uniformly operator-bounded sequence (a_i) to its Hilbert-ultrapower class is a trace-preserving isometry on the dense subspace $P^\omega \subset L^2(P^\omega)$. Its extension identifies $\mathcal{C} = L^2(P^\omega)$ with a closed subspace of $\mathcal{H}$.

For a bounded sequence $a = (a_i)$ in P , its coordinatewise left action on $\mathcal{H}$, and the adjoint action corresponding to (a_i^*) , preserve the bounded-algebra-sequence subspace of $\mathcal{C}$. Thus $\mathcal{C}$ reduces both actions. If $a \in \mathcal{I}_\omega$, then for every operator-bounded sequence (b_i) ,

$$\|a_i b_i\|_2 \leq \|b_i\| \|a_i\|_2 \longrightarrow_\omega 0.$$

It follows by density that the action on $\mathcal{C}$ factors through the tracial quotient and is precisely left multiplication by $[a_i]$. This factoring assertion is needed only on $\mathcal{C}$, not on the whole Hilbert ultrapower.

Let S^ω_{Hilb} be the coordinatewise action of the fixed bounded operator S on $\mathcal{H}$, and set

$$\widehat{S} = P_{\mathcal{C}} S^\omega_{\text{Hilb}}|_{\mathcal{C}}.$$

Since S commutes with every left multiplier from Q , its Hilbert ultrapower commutes with every bounded left Q -sequence. Compression to the reducing subspace $\mathcal{C}$ proves $[\widehat{S}, L_{Q^\omega}] = 0$.

Set $t = 1/(2m)$. We claim that

$$\begin{aligned} \|\widehat{S}, L_z\| &\leq 3\delta\|z\|, \\ e \in Q^\omega \text{ a projection, } \quad \tau(e) &= t, \quad z \in eP^\omega e. \end{aligned} \tag{37}$$

To lift e , take a bounded self-adjoint representative (a_i) in Q and set $f_i = 1_{[1/2, \infty)}(a_i)$. The scalar inequality $|x - 1_{[1/2, \infty)}(x)| \leq 2|x^2 - x|$ gives $\|f_i - a_i\|_2 \longrightarrow_\omega 0$, so $[f_i] = e$ and $\tau(f_i) \longrightarrow_\omega t$. The strict gap $t < 1/m$ allows us to replace f_i by zero off an ω -large set and assume $\tau(f_i) \leq 1/m$ for every i .

Given $z \in eP^\omega e$ and $\eta > 0$, the C^* -quotient norm gives a representative (b_i) with $\sup_i \|b_i\| \leq \|z\| + \eta$. The sequence $z_i = f_i b_i f_i$ represents z , lies in the required corners, and has the same norm bound. Consequently

$$\|[S, L_{z_i}]\| \leq 3\delta(\|z\| + \eta) \quad \text{for every } i.$$

This estimate passes to the Hilbert ultrapower. Its compression to $\mathcal{C}$ is $[\widehat{S}, L_z]$, since $\mathcal{C}$ reduces the bounded left sequence action. Letting η decrease to zero proves (37).

By Lemma 6.2, choose $D \subset Q^\omega$ free from the constant P , and put $\mathcal{L} = W^*(D, P) \subset P^\omega$. Faithfulness of the trace and the reduced-word inner products identify this algebra with the tracial

free product $D * P$. Let $G = P_{L^2(\mathcal{L})} \widehat{S}|_{L^2(\mathcal{L})}$. The subspace $L^2(\mathcal{L})$ reduces left multiplication by $\mathcal{L}$. Thus G commutes with L_D , and (37) gives the corner hypotheses of Theorem 3.1 with $A_0 = P$, $\epsilon = 3\delta$, and $t = 1/(2m)$.

Finally, every constant $L^2(P)$ -vector belongs to $\mathcal{C}$, by approximation by bounded vectors from P , and S_{Hilb}^ω sends the constant vector ξ to the constant vector $S\xi$. Both of these vectors also belong to $L^2(\mathcal{L})$. Therefore both compressions preserve the action of S on the constant subspace. As L_s preserves that subspace, for $\xi, \eta \in L^2(P)$ we have exactly

$$\langle [G, L_s] \xi, \eta \rangle = \langle [S, L_s] \xi, \eta \rangle.$$

Theorem 3.1, applied to the trace-zero symmetry $s \in P$, consequently yields on the original $L^2(P)$

$$\|[S, L_s]\| \leq 3c_1\delta.$$

Replacing Y_0 by Y costs at most $2\|Y_0 - Y\|\|s\| \leq 2\delta$, since Hilbert amplification preserves the difference norm. The initial block compression now gives the claimed bound $(3c_1 + 2)\delta$ for the original finite r, h . $\square$

7 From finite factors to arbitrary homomorphisms

We now remove the restrictions on the algebra and its representation, and then pass from inner commutators to arbitrary derivations. Throughout this section set

$$C_{\text{fac}} = \max\{3c_1 + 2, 1 + 2c_{\text{row}}, 2\}, \quad C = 3C_{\text{fac}} + 2. \quad (38)$$

These constants depend only on the numerical constants of the preceding sections.

7.1 Representations of factors

Proposition 7.1. *Let $P \subset \mathcal{B}(K)$ be a unital factor acting on a separable Hilbert space. For every $Y \in \mathcal{B}(K)$, every $h \geq 1$, and every $X \in M_h(P)$,*

$$\|[Y^{(h)}, X]\| \leq C_{\text{fac}} g_P(Y) \|X\|.$$

Proof. We first treat a type II_1 factor M in an arbitrary normal representation on a separable Hilbert space. Its predual is separable. Every cyclic normal representation embeds into the tracial standard representation. Indeed, if ξ is a cyclic vector, its normal positive functional has a trace density $a \in L^1(M)_+$ [14, Corollary 5.14 and Remark 5.15]. The map on cyclic orbits

$$x\xi \longmapsto xa^{1/2} \in L^2(M)$$

is isometric, because both squared norms equal the functional evaluated at x^*x . Its extension intertwines the representations and has a reducing range. Decomposing into orthogonal cyclic reducing subspaces, of which there are at most countably many, embeds the given representation into $\tilde{K} = L^2(M)^{\oplus \mathbb{N}}$. These are standard consequences of the normal representation theory of finite von Neumann algebras; see Takesaki [28].

Extend Y by zero off the embedded reducing subspace. The scalar commutator bound is unchanged. On the countable amplification, let p_n be the projection onto the first n copies. Since p_n commutes with the common left action,

$$[p_n Y p_n, L_x] = p_n [Y, L_x] p_n \quad (x \in M).$$

Proposition 6.3 applies to the compression of Y to $p_n \tilde{K}$, with constant $3c_1 + 2$. For any fixed finite matrix test, the corresponding compressed commutators converge strongly to the full commutator as $n \rightarrow \infty$. A uniform operator-norm bound passes to this strong limit. Restricting back to the original reducing subspace proves the assertion for type II_1 factors.

If P is properly infinite, choose isometries $v_1, \dots, v_h \in P$ with orthogonal ranges. The row $V = [v_1 \cdots v_h]$ satisfies $V^*V = I_{K^h}$. Apply the row and column derivation estimates of Section 2 to $x \mapsto [Y, x]$. With $\delta = g_P(Y)$, they give

$$\|YV - VY^{(h)}\| \leq c_{\text{row}}\delta, \quad \|Y^{(h)}V^* - V^*Y\| \leq c_{\text{row}}\delta.$$

For $X \in M_h(P)$, the identity

$$\begin{aligned} [Y, V XV^*] &= (YV - VY^{(h)})XV^* + V[Y^{(h)}, X]V^* \\ &\quad + VX(Y^{(h)}V^* - V^*Y) \end{aligned}$$

and $\|V XV^*\| = \|X\|$ imply

$$\|[Y^{(h)}, X]\| \leq (1 + 2c_{\text{row}})\delta\|X\|.$$

Here conjugation by the isometry $V : K^h \rightarrow K$ preserves the norm of the middle term, regardless of whether its range is all of K .

Finally, if P is a finite type I factor, average Y over its compact unitary group to obtain $Y_0 \in P'$ with $\|Y - Y_0\| \leq \delta$. Then

$$[Y^{(h)}, X] = [(Y - Y_0)^{(h)}, X],$$

which gives the constant 2. These three cases exhaust the factors and prove the proposition. $\square$

7.2 Central decomposition

Proposition 7.2. *Let $P \subset \mathcal{B}(K)$ be any unital von Neumann algebra on a separable Hilbert space. Then*

$$\|[Y^{(h)}, X]\| \leq C g_P(Y) \|X\| \quad (Y \in \mathcal{B}(K), X \in M_h(P), h \geq 1).$$

Proof. Write $\delta = g_P(Y)$. Average Y over the compact unitary groups of the finite-dimensional algebras generated by finite families of central projections. Direct these families by inclusion and take a cofinal weak operator cluster limit. The resulting operator Y_1 commutes with every central projection, hence with $Z(P)$, and satisfies

$$\|Y_1 - Y\| \leq \delta, \quad g_P(Y_1) \leq 3\delta. \quad (39)$$

For the first inequality, each unitary being averaged belongs to P , so its conjugation changes Y by at most δ . The inequality is preserved by averages and weak operator limits. The second follows from $g_P(Y_1) \leq g_P(Y) + 2\|Y_1 - Y\|$.

Use the central direct-integral decomposition on a separable Hilbert space [28]; precise section locators for the factor fibers, decomposable operators, and norm formula are Blackadar [1, Sections III.1.6.1–4 and III.5.1.10–15]:

$$K = \int^{\oplus} K_{\alpha} d\mu(\alpha), \quad P = \int^{\oplus} P_{\alpha} d\mu(\alpha),$$

where the P_{α} are factors almost everywhere and the fibers are separable. The center acts as the scalar diagonal algebra, so its commutant consists of the decomposable operators. In particular, Y_1 has fibers $Y_1(\alpha)$. We verify carefully that

$$g_{P_{\alpha}}(Y_1(\alpha)) \leq 3\delta \quad \text{for almost every } \alpha. \quad (40)$$

The separable direct-integral decomposition admits countably many bounded measurable sections whose values generate P_α almost everywhere; fix such a family. Take all rational complex $*$ -polynomials in these sections, including the rational scalar constants. For each such section $b \in P$, form the contraction

$$c_b = b(1 + b^*b)^{-1/2} \in P.$$

On every fiber where the generators have the stated property, these contractions are norm dense in the contraction ball of the generated C^* -algebra. To see this, the polynomials are norm dense, the displayed transformation is norm continuous, and its range contains the open unit ball: its inverse there is

$$c \mapsto c(1 - c^*c)^{-1/2}.$$

The closed unit ball is the norm closure of the open one. Kaplansky density then makes these countably many fiber contractions strongly dense in the unit ball of P_α .

For each c_b , the bound $\|[Y_1, c_b]\| \leq 3\delta$ implies the same fiber bound outside a null set, by the essential-supremum norm formula. Removing the union of these countably many null sets leaves a single conull set on which every test holds. Strong approximation by the contraction tests passes the bound to every element of the fiber unit ball, proving (40).

Apply Proposition 7.1 on that conull set. For any fixed $X \in M_h(P)$, decomposability of its entries and the same norm formula give

$$\|[Y_1^{(h)}, X]\| \leq 3C_{\text{fac}}\delta\|X\|.$$

Restoring Y costs at most $2\delta\|X\|$ by (39), giving $C = 3C_{\text{fac}} + 2$. □

7.3 Removing separability

Proof of Theorem 1.2. Let $P \subset \mathcal{B}(K)$ be arbitrary, and fix $h \geq 1$, $X \in M_h(P)$, and a unit vector $\xi = (\xi_1, \dots, \xi_h) \in K^h$. Let $A_X \subset P$ be the unital C^* -algebra generated by the entries of X . The C^* -algebra in $\mathcal{B}(K)$ generated by these entries and Y is norm separable. The closed span of its orbits of the finitely many ξ_i is therefore a separable common reducing subspace K_0 for A_X and Y . Denote restriction of A_X to K_0 by σ , put $Y_0 = Y|_{K_0}$, and set $P_0 = \sigma(A_X)''$.

For $a \in A_X$,

$$\|[Y_0, \sigma(a)]\| \leq g_P(Y)\|a\|.$$

The C^* -quotient norm of the image gives, for every $b \in \sigma(A_X)$,

$$\|[Y_0, b]\| \leq g_P(Y) \inf_{\sigma(a)=b} \|a\| = g_P(Y)\|b\|.$$

By Kaplansky density the same scalar commutator bound holds on P_0 . Proposition 7.2, applied to P_0 and $\sigma_h(X)$, now yields

$$\|[Y_0^{(h)}, \sigma_h(X)]\| \leq Cg_P(Y)\|\sigma_h(X)\| \leq Cg_P(Y)\|X\|.$$

Since K_0 reduces all the operators in this test and contains every component of ξ , the restricted commutator on ξ equals the original one. Taking the supremum over unit test vectors proves (1). If $K = \{0\}$, the estimate is vacuous. □

7.4 Derivations and similarity

Theorem 1.2 now controls inner commutators in every representation. To reach Kirchberg's criterion we must implement an arbitrary bounded derivation. Cyclic-domain implementation supplies finitely many columns at a time; the uniform commutator estimate lets us pass from these finite collections to the full derivation.

Proposition 7.3. *Let A be a unital C^* -algebra and $\sigma : A \rightarrow \mathcal{B}(K)$ a unital $*$ -representation. Every bounded σ -derivation $\Delta : A \rightarrow \mathcal{B}(K)$ satisfies*

$$\|\Delta\|_{\text{cb}} \leq C\|\Delta\|$$

and is implemented by a bounded operator on K .

Proof. Decompose K into cyclic reducing subspaces for σ :

$$K = \bigoplus_{i \in I} K_i.$$

Such a decomposition is obtained by taking a maximal orthogonal family of cyclic reducing subspaces; a nonzero vector in its orthogonal complement would generate another one. The index set I can have any cardinality. Let q_i denote the projection onto K_i .

The map $a \mapsto \Delta(a)|_{K_i}$ is a rectangular derivation with cyclic domain representation $\sigma|_{K_i}$ and full output representation σ . Lemma 2.4 gives a bounded map $V_i : K_i \rightarrow K$ such that

$$\Delta(a)|_{K_i} = V_i \sigma(a)|_{K_i} - \sigma(a) V_i.$$

For a finite subset $F \subset I$, put

$$q_F = \sum_{i \in F} q_i, \quad V_F = \sum_{i \in F} V_i q_i \in \mathcal{B}(K).$$

Every V_F is bounded, and the reducing property gives the exact identity

$$[V_F, \sigma(a)] = \Delta(a) q_F \quad (a \in A). \quad (41)$$

Only the scalar commutator norm, rather than the norm of V_F , needs to be uniform in F . Indeed, taking the infimum over lifts of an element $b \in \sigma(A)$ in (41) gives

$$\|[V_F, b]\| \leq \|\Delta\| \|b\|.$$

This uses the C^* -quotient norm of the $*$ -image and does not require σ to be faithful. Kaplansky density then implies

$$g_{\sigma(A)''}(V_F) \leq \|\Delta\|.$$

Apply Theorem 1.2 to V_F and $\sigma(A)''$. For every $X \in M_h(A)$,

$$\begin{aligned} \|\Delta_h(X) q_F^{(h)}\| &= \|[V_F^{(h)}, \sigma_h(X)]\| \\ &\leq C\|\Delta\| \|\sigma_h(X)\| \leq C\|\Delta\| \|X\|. \end{aligned}$$

The projections q_F converge strongly to the identity as a net indexed by finite subsets of I . Hence the displayed bound passes to $\Delta_h(X)$ for every fixed h and X . This proves the claimed complete boundedness. Lemma 2.2 then supplies a bounded implementing operator. $\square$

Proof of Theorem 1.1. For every faithful unital $*$ -representation of A , each bounded derivation into the ambient bounded operators is inner by Proposition 7.3. Thus condition (2) of Theorem 2.5 holds for A . Its equivalent similarity property gives, for every bounded unital homomorphism $\pi : A \rightarrow \mathcal{B}(H)$, a bounded invertible S with bounded inverse such that $S\pi(\cdot)S^{-1}$ is a $*$ -homomorphism. The identity isomorphism treats the zero Hilbert space as well. $\square$

7.5 Hyperreflexivity

The uniform commutator estimate also controls distance to any von Neumann algebra. We apply it to the commutant, whose projections describe the algebra's invariant subspaces.

Proof of Corollary 1.3. Put $P = M'$. Since M is self-adjoint, its invariant subspaces are reducing and their orthogonal projections are exactly the projections in P . Hence α_M is the usual off-diagonal distance seminorm. For every projection $e \in P$, the two off-diagonal blocks give

$$\|[T, e]\| = \max\{\|(1 - e)Te\|, \|eT(1 - e)\|\} \leq \alpha_M(T),$$

where the last inequality also uses the projection $1 - e$. Spectral step approximations of a self-adjoint contraction $a \in P$ express it, modulo a scalar, as a norm limit of linear combinations of projections in P whose coefficients have total absolute value at most two. Thus $\|[T, a]\| \leq 2\alpha_M(T)$. The real and imaginary parts of any contraction in P are self-adjoint contractions, so $g_P(T) \leq 4\alpha_M(T)$.

Arveson's commutant-distance formula, in the form recalled in Christensen et al. [7, Equation (2.1)], gives

$$2 \operatorname{dist}(T, M) = \|\operatorname{ad}_T|_P\|_{\text{cb}}, \quad \operatorname{ad}_T(a) = [T, a],$$

since $P' = M$. Taking the supremum over finite matrix sizes in Theorem 1.2 therefore yields

$$2 \operatorname{dist}(T, M) \leq C g_P(T) \leq 4C \alpha_M(T),$$

which proves Equation (2). □

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